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EarthBridge: A Solution for 4th Multi-modal Aerial View Image Challenge Translation Track

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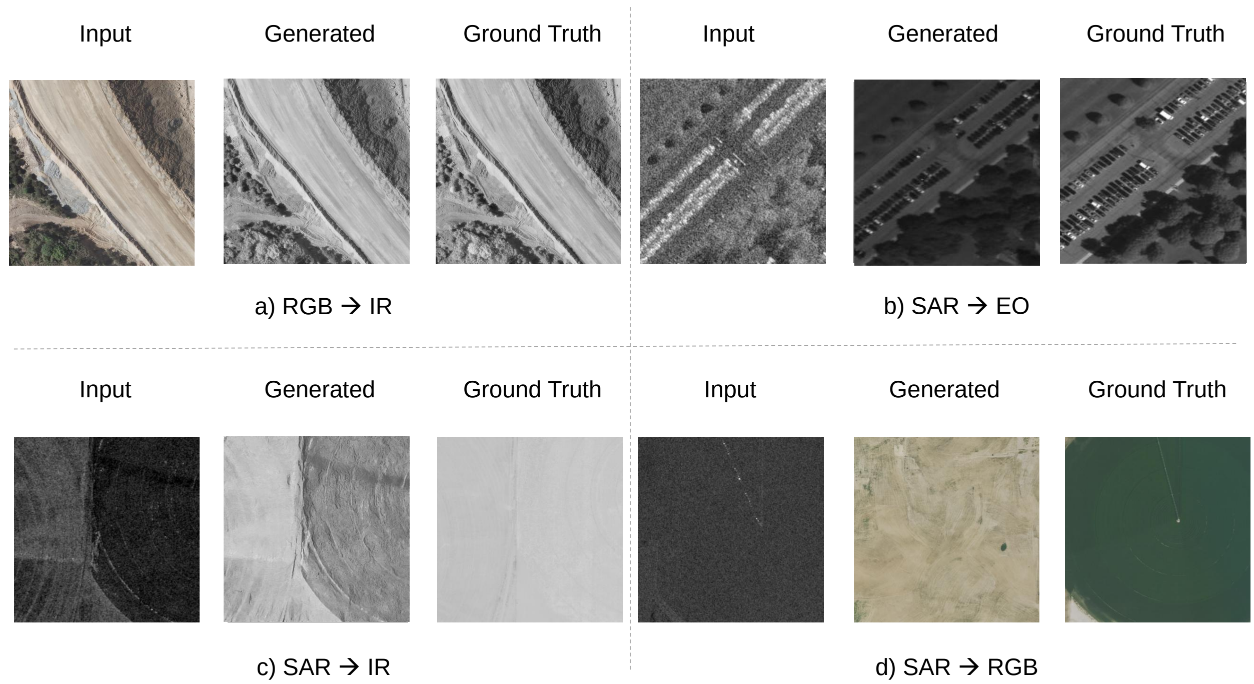


Figure 1. Teaser: The proposed EarthBridge framework for multi-modal aerial-view image translation across Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR) modalities. Our method accurately bridges the domain gap, preserving structural integrity while synthesizing high-fidelity target textures.

Abstract

*Cross-modal image-to-image translation among Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR) sensors is essential for comprehensive multi-modal aerial-view analysis. However, translating between these modalities is notoriously difficult due to their distinct electromagnetic signatures and geometric characteristics. This paper presents **EarthBridge**, a high-fidelity translation framework developed for the 4th Multi-modal Aerial View Image Challenge – Translation (MAVIC-T). We explore two distinct methodologies: **Diffusion Bridge Implicit Models (DBIM)**, which we generalize using non-Markovian bridge processes for high-quality deterministic sampling, and **Contrastive Unpaired Translation (CUT)**, which utilizes contrastive learning for structural consistency. Our EarthBridge framework employs a channel-concatenated UNet denoiser trained with Karras-weighted bridge scalings and a specialized "booting noise" initialization to handle the inherent ambiguity in cross-modal mappings. We evaluate these methods across all four challenge tasks (SAR→EO, SAR→RGB, SAR→IR, RGB→IR), achieving superior spatial detail and spectral accuracy. Our solution achieved a composite score of 0.38, securing the second po-*

sition on the MAVIC-T leaderboard.

1. Introduction

The Multi-modal Aerial View Image Challenge for cross-domain image translation [1, 19, 20] is a leading challenge held in conjunction with the Perception Beyond the Visible Spectrum (PBVS) workshop from 2023 to 2025. Data collected across multiple sensing modalities—such as Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR)—provide complementary perspectives on the same scene. However, data coverage limitations and modality-specific constraints often hinder the full utilization of these diverse sources. For instance, while SAR sensors offer all-weather capabilities and can penetrate atmospheric obstructions, their imagery is complex to interpret, and publicly available datasets are scarce. Conversely, widely available EO imagery is limited by lighting conditions and atmospheric visibility. Domain translation seeks to bridge this gap by converting one modality into another, enabling cross-modal data synthesis and enhancing sensor interoperability. By focusing on translation tasks such as RGB→IR, SAR→EO, SAR→IR, and SAR→RGB, the MAVIC-T challenge aims to bolster existing data volume, increase data diversity across regions, and serve as a platform for advancing conditioned image synthesis techniques, ultimately fostering the development of more adaptable and robust models for multi-modal image analysis.

SAR images possess unique attributes that present challenges for both human observers and vision artificial intelligence (AI) models to interpret, owing to their electromagnetic (EM) characteristics [10–13, 21].

In this paper, we propose **EarthBridge**, a multi-method framework for cross-modal aerial image translation. Our primary approach is built on Diffusion Bridge Implicit Models (DBIM) [43], which we generalize with non-Markovian bridge processes to enable efficient deterministic sampling. In parallel, we explore Contrastive Unpaired Translation (CUT) [24] as a powerful contrastive-learning-based alternative. Unlike standard conditional diffusion models that diffuse targets to pure noise and rely on classifier-free guidance or simple concatenation for the reverse path, EarthBridge implements a genuine stochastic diffusion bridge. By directly constraining the forward generative trajectory between the source modality x_T and target modality x_0, and parameterizing the UNet denoiser to solve the bridge marginals under Karras-weighted scalings [15], EarthBridge (DBIM) achieves high-fidelity translation across disparate modalities. We utilize pixel-level modeling throughout both approaches to preserve the full spectral and spatial resolution of the remote sensing data.

Our findings indicate that the deterministic DBIM sampling significantly improves inference speed, achieving high-quality results in as few as 5 sampling steps in RGB→IR. Qualitatively, EarthBridge demonstrates a superior ability to reconstruct complex urban layouts and fine-grained textures in challenging tasks such as SAR-to-RGB, effectively bridging the domain gap. Quantitatively, our solution achieves a composite task score of 0.269 for the SAR→EO task and 0.38 overall, ultimately placing second in the MAVIC-T challenge leaderboard.

Our main contributions are summarized as follows:

- We propose **EarthBridge**, a multi-method framework for paired cross-modal aerial image translation incorporating both Diffusion Bridge Implicit Models (DBIM) and Contrastive Unpaired Translation (CUT), providing a robust comparison between diffusion-based and contrastive-learning-based approaches.
- We develop a specialized training protocol using Karras-weighted bridge scalings and pixel-level modeling, optimized for the high dynamic range and resolution requirements of multimodal remote sensing data.
- We conduct comprehensive experiments on all four MAVIC-T tasks (SAR→EO, SAR→RGB, SAR→IR, RGB→IR), reporting competitive results and providing detailed quantitative and qualitative analysis.

2. Related Work

Translating across remote sensing (RS) modalities, such as SAR to optical (S2O), is critical for multisource data fusion. In this section, we provide a comprehensive literature review of image-to-image translation, diffusion bridge models, and cross-modality translation to complement the specialized discussion on remote sensing.

General Image-to-Image Translation. Image-to-image translation has been extensively studied in computer vision. Foundational works include Pix2Pix for paired conditional translation [14] and CycleGAN for unpaired translation [45]. Subsequent advances include contrastive learning for unpaired translation [24], zero-shot [25] and one-step translation with text-to-image models [26].

Diffusion Bridge Models. Unlike standard diffusion models that generate from noise, diffusion bridge models learn a stochastic process that directly connects source and target distributions, making them well-suited for paired image-to-image translation. Denoising Diffusion Bridge Models (DDBM) [44] define a variance-preserving bridge between data distributions. Related formulations include Brownian Bridge Diffusion Models (BBDM) [16] and its bidirectional extension BiBBDM [40], Image-to-Image Schrödinger Bridge (I^2SB) [17], Diffusion Schrödinger Bridge matching [34], Dual Diffusion Implicit Bridges [35], and Direct Diffusion Bridge with data consistency [2]. Recent extensions include Consistency Diffusion Bridge [6], Diffusion Bridge Implicit Models [43], deterministic translation via denoising Brownian bridge with dual approximators [39], and adaptive domain shift for cross-modality translation [38].

Cross-Modality Translation. Image translation across diverse sensor modalities—grayscale, near-infrared (NIR), thermal colorization, and color transfer—has been addressed with probabilistic color correction [23], NIR imagery colorization [36], deep colorization [21, 41].

GAN-based RS Translation. Early attempts at translating between SAR and optical modalities leveraged Generative Adversarial Networks (GANs). For instance, Wang et al. [37] proposed a hierarchical latent feature approach to improve the fidelity of SAR-to-optical translation. Hu et al. [9] specifically addressed this task in fire-disturbed regions using GANs to reconstruct optical details from SAR signals, which is vital for post-fire damage assessment when clouds or smoke obscure optical sensors.

Diffusion Models for RS Translation. Reflecting the recent shift in generative modeling, diffusion models have been increasingly applied to RS translation tasks. Qin et al. [28] introduced a conditional diffusion model incorporating spatial-frequency refinement to capture both global structure and fine-grained textures in SAR-to-optical translation. To address the computational overhead of iterative sampling, Qin et al. [29] later developed an efficient end-to-end diffusion model capable of one-step SAR-to-optical translation, significantly reducing inference latency while maintaining high generation quality. Furthermore, Wang et al. [38] proposed adaptive domain shift mechanisms within diffusion frameworks to better handle the large distribution gaps inherent in cross-modality remote sensing datasets. These advancements underscore the growing role of stochastic bridge processes and diffusion-based refinement in enabling robust, high-resolution translation across heterogeneous sensor modalities.

3. Preliminary

Problem Formulation Let $\mathcal{X} \subset \mathbb{R}^{C_s \times H \times W}$ denote the source domain and $\mathcal{Y} \subset \mathbb{R}^{C_t \times H \times W}$ denote the target domain, where C_s and C_t are the number of channels in the source and target modalities, respectively. Given a set of spatially-aligned training pairs $\{(x_i, y_i)\}_{i=1}^{N}$ with $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, the goal of cross-modal image-to-image translation is to learn a mapping $G : \mathcal{X} \to \mathcal{Y}$ that minimizes a composite distance between the generated image $\hat{y} = G(x)$ and the ground-truth target y. In the context of the MAVIC-T challenge, four translation tasks are defined (see Tab. 1): SAR→EO ($1 \to 1$ channel, 256×256), SAR→RGB ($1 \to 3$, 1024×1024), SAR→IR ($1 \to 1$, 1024×1024), and RGB→IR ($3 \to 1$, 1024×1024).

Evaluation Metric The MAVIC-T challenge evaluates submissions using a composite score derived from three complementary metrics: LPIPS [42] (Learned Perceptual Image Patch Similarity, based on VGG-16) for perceptual fidelity, FID [8] (Fréchet Inception Distance, based on InceptionV3) for distributional similarity, and L1 norm for pixel-level structural accuracy. These are combined into a task score:

$$S_{\text{task}} = \frac{\frac{2}{\pi}\arctan(\text{FID}) + \text{LPIPS} + \text{L1}}{3}, \tag{1}$$

where the $\arctan$ normalization maps FID into the $[0, 1]$ range to balance its contribution against the other two metrics. The overall score is the average of the four task scores, with a penalty of $+1$ added for each unattempted domain, encouraging participants to develop generalizable solutions across all modality pairs.

4. EarthBridge: Multi-method Framework for RS Translation

We propose EarthBridge, a flexible framework that explores two distinct methodologies for cross-modal image translation: Diffusion Bridge Implicit Models (DBIM) and Contrastive Unpaired Translation (CUT).

4.1. Method 1: Diffusion Bridge Implicit Models (DBIM)

Diffusion models have recently revolutionized generative modeling, yet they are fundamentally designed to transport complex distributions to a standard Gaussian prior. Adapting them to translate between two arbitrary distributions, such as heterogenous remote sensing modalities, requires conditioning the diffusion process on a fixed endpoint. This is achieved via Doob's h-transform [5, 30].

Stochastic Bridges via h-transform. Given a base diffusion process $dx_t = f(x_t, t)dt + g(t)dw_t$, the conditioned SDE that almost surely reaches a prescribed target $x_0 = y$ at $t = 0$ starting from a source $x_T = x$ at $t = T$ is given by:

$$\begin{aligned} dx_t &= f(x_t, t)dt + g(t)dw_t \\ &\quad + g(t)^2 \nabla_{x_t} \log p(x_0 = y \mid x_t)dt, \end{aligned} \tag{2}$$

where $p(x_0 = y \mid x_t)$ is the transition density of the original process. When the endpoints are fixed, the resulting conditioned process is called a *diffusion bridge* [3, 7, 18, 27, 32, 33], which forms the foundational mechanism for transporting one state to another. In the context of cross-modal translation, x represents the source modality (e.g., SAR) and y the target modality (e.g., optical). This formulation ensures that the generative trajectory is directly constrained by both endpoints, providing a more principled framework for paired translation than standard conditional diffusion.

Denoising Diffusion Bridge Models. Denoising Diffusion Bridge Models (DDBM) [44] provide a tractable realization of Eq. 2. Under a variance-preserving (VP) schedule, the forward bridge process constructs intermediate states z_t as:

$$z_t = a_t x + b_t y + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{3}$$

where a_t, b_t, σ_t are schedule-dependent coefficients. While DDBM excels in high-fidelity generation, its inference typically relies on simulating Eq. 2 via computationally expensive numerical solvers (e.g., Heun or stochastic samplers), often requiring hundreds of function evaluations (NFE) for optimal results. In this work, we adopt the more efficient *implicit* formulation to accelerate this process.

4.2. EarthBridge: Non-Markovian DBIM

We present our DBIM approach which generalizes the diffusion bridge framework with non-Markovian bridge processes defined on discretized timesteps. Our solution leverages the deterministic sampling property of DBIM to achieve high-fidelity cross-modal translation with significantly fewer sampling steps compared to the vanilla DDBM. Fig. 2 shows an overview of our EarthBridge pipeline.

Non-Markovian Diffusion Bridges. In contrast to the continuous-time Markovian bridge processes in DDBM, we consider a family of non-Markovian diffusion bridges defined over a sequence of discretized timesteps $0 = t_0 < t_1 < \cdots < t_N = T$, controlled by a variance parameter $\rho \in \mathbb{R}^{N-1}$. The UNet denoiser is trained to predict the bridge marginals governed by the schedule-dependent coefficients (a_t, b_t, c_t), where the source image $\boldsymbol{x}$ is supplied to the network via channel concatenation. This concatenation is strictly an architectural mechanism to inject the mandatory endpoint parameter $\boldsymbol{x}$ into the SDE formulation, not an ad-hoc conditioning trick akin to standard image-to-image models. The joint probability distribution $q^{(\rho)}(\boldsymbol{z}_{t_{0:N-1}} \mid \boldsymbol{x})$ is factorized through a chain of conditional distributions:

$$q^{(\rho)}(\boldsymbol{z}_{t_{0:N-1}} \mid \boldsymbol{x}) = q_0(\boldsymbol{z}_{t_0}) \prod_{n=1}^{N-1} q^{(\rho)}(\boldsymbol{z}_{t_n} \mid \boldsymbol{z}_0, \boldsymbol{z}_{t_{n+1}}, \boldsymbol{x}), \tag{4}$$

where q_0 is the target modality distribution and each step $q^{(\rho)}$ is defined as:

$$\mathcal{N}\big(a_{t_n}\boldsymbol{x} + b_{t_n}\boldsymbol{z}_0 + \sqrt{c_{t_n}^2 - \rho_n^2}\,\boldsymbol{\eta}_{n+1},\, \rho_n^2 \boldsymbol{I}\big), \tag{5}$$

with $\boldsymbol{\eta}_{n+1} = (\boldsymbol{z}_{t_{n+1}} - a_{t_{n+1}}\boldsymbol{x} - b_{t_{n+1}}\boldsymbol{z}_0)/c_{t_{n+1}}$. Here, a_t, b_t, c_t are known coefficients determined by the VP log-SNR schedule (Sec. 4.1). Under this construction, the marginal distributions $q^{(\rho)}(\boldsymbol{z}_{t_n} \mid \boldsymbol{x})$ are preserved to be consistent with the continuous-time forward bridge (Eq. (3)). The variance parameter ρ controls the level of stochasticity in the sampling process, ranging from a purely Markovian stochastic process to a fully deterministic implicit model.

Reverse Generative Process and Sampling. The generative process is initialized at the source image $\boldsymbol{z}_T = \boldsymbol{x}$ and iteratively samples $\boldsymbol{z}_{t_n}$ from $\boldsymbol{z}_{t_{n+1}}$ via a parameterized data predictor $D_\theta(\boldsymbol{z}_{t_{n+1}}, t_{n+1}, \boldsymbol{x})$. The core updating rule is given by:

$$\begin{aligned} \boldsymbol{z}_{t_n} &= a_{t_n}\boldsymbol{x} + b_{t_n}\hat{\boldsymbol{z}}_0 \\ &\quad + \sqrt{c_{t_n}^2 - \rho_n^2}\,\hat{\boldsymbol{\epsilon}} + \rho_n \boldsymbol{\epsilon}, \end{aligned} \tag{6}$$

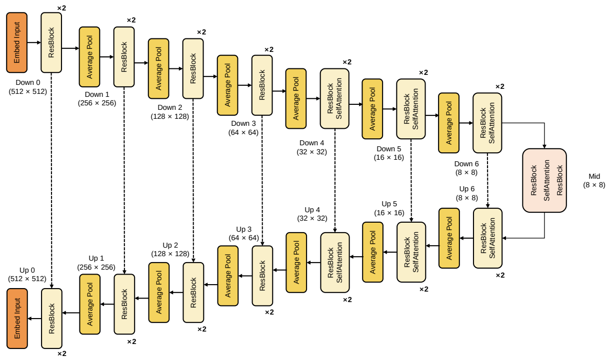
where $\hat{\boldsymbol{z}}_0 = D_\theta(\boldsymbol{z}_{t_{n+1}}, t_{n+1}, \boldsymbol{x})$ is the clean data prediction and $\hat{\boldsymbol{\epsilon}}$ is the predicted noise at time t_{n+1}. Setting $\rho_n = 0$ for all steps yields the deterministic *Diffusion Bridge Implicit Model (DBIM)*, which can be viewed as an Euler discretization of a probability flow ODE specifically tailored for diffusion bridges.

Booting Noise. A critical refinement in our sampling procedure is the introduction of *booting noise* at the initial step ($t_N = T$). Since $c_{t_N} = 0$, the denominator in $\hat{\boldsymbol{\epsilon}}$ leads to singularity. To address this, we force the initial step to be stochastic by choosing $\rho_{N-1} = c_{t_{N-1}}$, which injects a standard Gaussian booting noise $\boldsymbol{\epsilon}_{\text{boot}}$. This booting noise serves as the latent variable, encoding the one-to-many mapping ambiguity in RS translation (e.g., several plausible infrared images for a single SAR input).

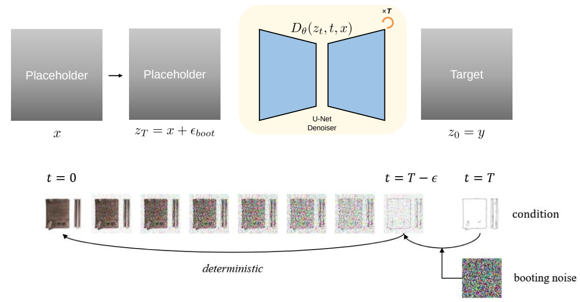
Channel Handling for Mismatched Modalities. For tasks where source and target modalities have different channel counts (e.g., RGB→IR or SAR→RGB), we operate the bridge in a shared `model_channels` space (typically 3 for RGB-related tasks). Single-band modalities are expanded to `model_channels` by channel repetition before entering the UNet, ensuring the concatenation $[\boldsymbol{z}_t, \boldsymbol{x}]$ remains well-defined across all tasks.

4.3. Method 2: Contrastive Unpaired Translation (CUT)

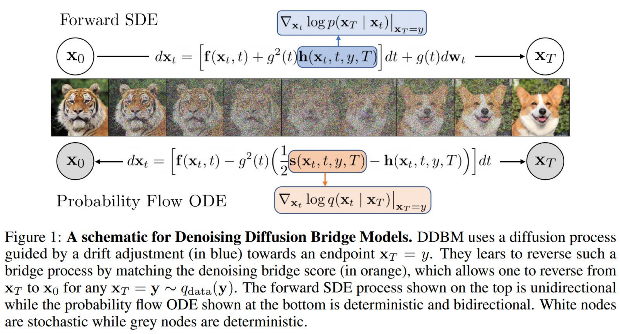
In addition to our diffusion-based approach, we leverage Contrastive Unpaired Translation (CUT) [24] as an alternative method for cross-modal translation. Unlike diffusion



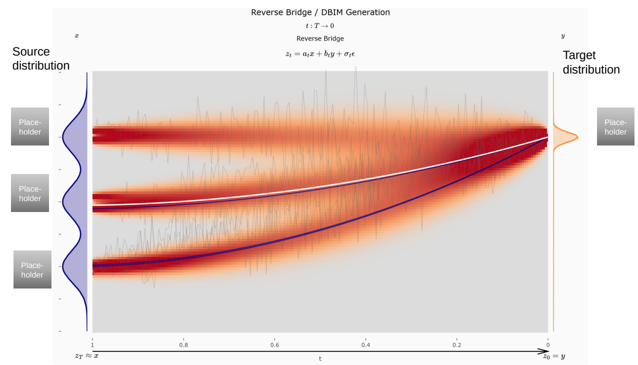
(a) UNet denoiser for 1024px tasks.



(b) **[TODO: Subfigure placeholder.]**



(c) **[TODO: Subfigure placeholder.]**



(d) **[TODO: Subfigure placeholder.]**

Figure 2. UNet architecture of our DBIM for 1024px tasks (SAR→RGB, RGB→IR). We train at 512×512 and infer at full resolution. Source $\boldsymbol{x}$ is channel-concatenated with noisy sample $\boldsymbol{z}_t$ and processed by this denoiser. Bridge scalings a_t, b_t, c_t construct $\boldsymbol{z}_t$ during training; deterministic DBIM sampling generates $\hat{\boldsymbol{z}}_0$ during inference.

bridges, CUT utilizes contrastive learning to maintain structural correspondence. Our implementation uses a Patch-based Contrastive Learning (PCL) strategy to maximize the mutual information between corresponding patches of the source and generated target images.

5. Experiments

5.1. Dataset and Preprocessing

We use the MAVIC-T 2025 dataset from the challenge organizers, with imagery from UNICORN, USGS EROS HRO, and UMBRA across four regions: Bingham, Centerfield, Manhattan, and UC Davis. Tab. 1 summarizes the four translation tasks. SAR→EO operates at 256×256; the other three tasks use 1024×1024. For SAR→IR and SAR→RGB, we spatially align SAR scenes with optical mosaics before resampling. We also create crop-augmented training sets for the 1024×1024 tasks to increase data diversity.

5.2. Architecture and Training Objectives

UNet Denoising Backbone Our denoiser D_θ is a UNet [31] that predicts the clean target from the noisy sample $\boldsymbol{z}_t$ conditioned on timestep t and source $\boldsymbol{x}$. We use task-specific scaling (128 base channels with resolution-dependent multipliers) and train with the Prodigy [22] optimizer. Detailed architectural parameters and source conditioning strategies are provided in Sec. 6.1.

Table 1. Task specifications and training data for the MAVIC-T challenge. Crop augmentation uses 1024×1024 sliding windows with stride 512 (50% overlap); SAR→optical pairs are georeference-aligned.

Task	Resolution	Channels	Samples (Orig.)	Samples (Aug.)
SAR→EO	256×256	1→1	89,411	–
SAR→RGB	1024×1024	1→3	10,576	75,703
SAR→IR	1024×1024	1→1	10,576	74,317
RGB→IR	1024×1024	3→1	2,272	7,388

Training Objective The primary training objective is a weighted mean squared error between the denoised prediction $D_\theta(\boldsymbol{z}_t, t, \boldsymbol{x})$ and the clean target $\boldsymbol{y}$:

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y}),\, t,\, \epsilon}\left[w(t) \cdot \|D_\theta(\boldsymbol{z}_t, t, \boldsymbol{x}) - \boldsymbol{y}\|_2^2\right], \quad (7)$$

Table 2. Main results on the MAVIC-T evaluation set. ↓ lower is better.

Track	Method	FID↓	LPIPS↓	L1↓	Score↓	NFE	Time
SAR→EO	DBIM 500	0.222	0.504	0.080	0.269	1000	0.42 s
SAR→RGB	DBIM 1000	0.878	0.636	0.214	0.576	2000	160 s
SAR→IR	CUT huge	0.645	0.603	0.145	0.464	1	0.47 s
RGB→IR	DBIM 5	0.357	0.151	0.090	0.200	10	1.5 s

where $w(t)$ is the Karras bridge weighting. The diffusion time t is sampled from a Karras inverse-ρ distribution. During inference, we utilize deterministic DBIM sampling for efficiency. Full technical details on the loss weighting, time-sampling distribution, and per-task training configurations are deferred to the appendix (Sec. 6).

Baseline: Contrastive Unpaired Translation (CUT) To assess the performance of our EarthBridge framework, we incorporate Contrastive Unpaired Translation (CUT) [24] as a primary baseline. Our implementation of CUT follows the original patch-based contrastive learning strategy, where patches from the generated target image are encouraged to be closer to the corresponding patches from the source image in a shared embedding space. We adapt the CUT architecture with ResNet-based generators and PatchGAN discriminators, scaled to match the resolution requirements of each task (e.g., medium tier for 256×256 and large/huge tiers for 1024×1024 as detailed in Sec. 6).

5.3. Quantitative Results

Tab. 2 presents the quantitative performance comparison between EarthBridge and the CUT baseline across the challenge tasks. The composite task score provides a balanced evaluation of perceptual, distributional, and structural fidelity. EarthBridge demonstrates consistent performance across both low- and high-resolution tasks.

5.4. Qualitative Results

Fig. 3 shows qualitative examples across all four tasks. EarthBridge preserves fine-grained structural information from the source modality while accurately synthesizing target textures, demonstrating consistent performance across both low- and high-resolution settings.

5.5. Failure Case and Challenging Scenario Analysis

[TODO: Add failure case analysis and challenging scenarios.]

6. Implementation Details

This section provides the comprehensive technical details of the EarthBridge architecture, training hyperparameters, and multi-modal refinement modules.

Table 3. DBIM (SAR→EO, SAR→RGB, RGB→IR).

Parameter	SAR→EO	SAR→RGB	RGB→IR
Base channels	64	96	128
Channel multipliers	(1,2,3,4)	(1,1,2,2,4,4)	
Attention resolutions	N/A	32, 16, 8	
Training resolution	256^2	512^2	
Epochs	55	24	86

Table 4. CUT (SAR→IR). ResNet-9block, 4-layer PatchGAN.

Parameter	SAR→IR
ngf / ndf	256
Generator	ResNet-9block
Discriminator	4-layer PatchGAN
Train res.	512^2
Epochs	11

6.1. Architecture, Sampling, and Scheduler Settings

The EarthBridge architecture and sampling configurations are designed for high-fidelity cross-modal translation with efficiency. We utilize a `UNet2DModel` backbone (ADM-style [4]) with per-task architectural parameters: number of channels, number of residual blocks, attention resolutions, channel multipliers, resolution, and training settings (epochs, augmentations) as detailed in Tab. 3. Timestep information is injected via sinusoidal embeddings and Adaptive Group Normalization (AdaGN). Source image conditioning is implemented via channel-wise concatenation (`concat` mode) $[\boldsymbol{z}_t, \boldsymbol{x}]$.

Sampling is performed using the deterministic DBIM formulation [43] with task-dependent inference steps (see Tab. 2) and a Gaussian booting noise ϵ_{boot} at $t = T$. The bridge process follows a variance-preserving (VP) log-SNR schedule with $\beta_d = 2.0$ and $\beta_{\min} = 0.1$. For training, we sample diffusion time from a Karras inverse-ρ distribution with $\rho = 7$. Per-task architectural and training configurations are summarized in Tab. 3.

The CUT baseline follows the standard configuration with ResNet-based generators and PatchGAN discriminators. We use the *medium* tier for the 256 × 256 SAR→EO task and the *huge* tier for the 1024 × 1024 tasks to ensure sufficient capacity for high-resolution synthesis. The training hyperparameters for CUT, including the number of filters and residual blocks, are detailed in Tab. 4.

We train EarthBridge using the Prodigy adaptive optimizer across all translation tasks. The training objective

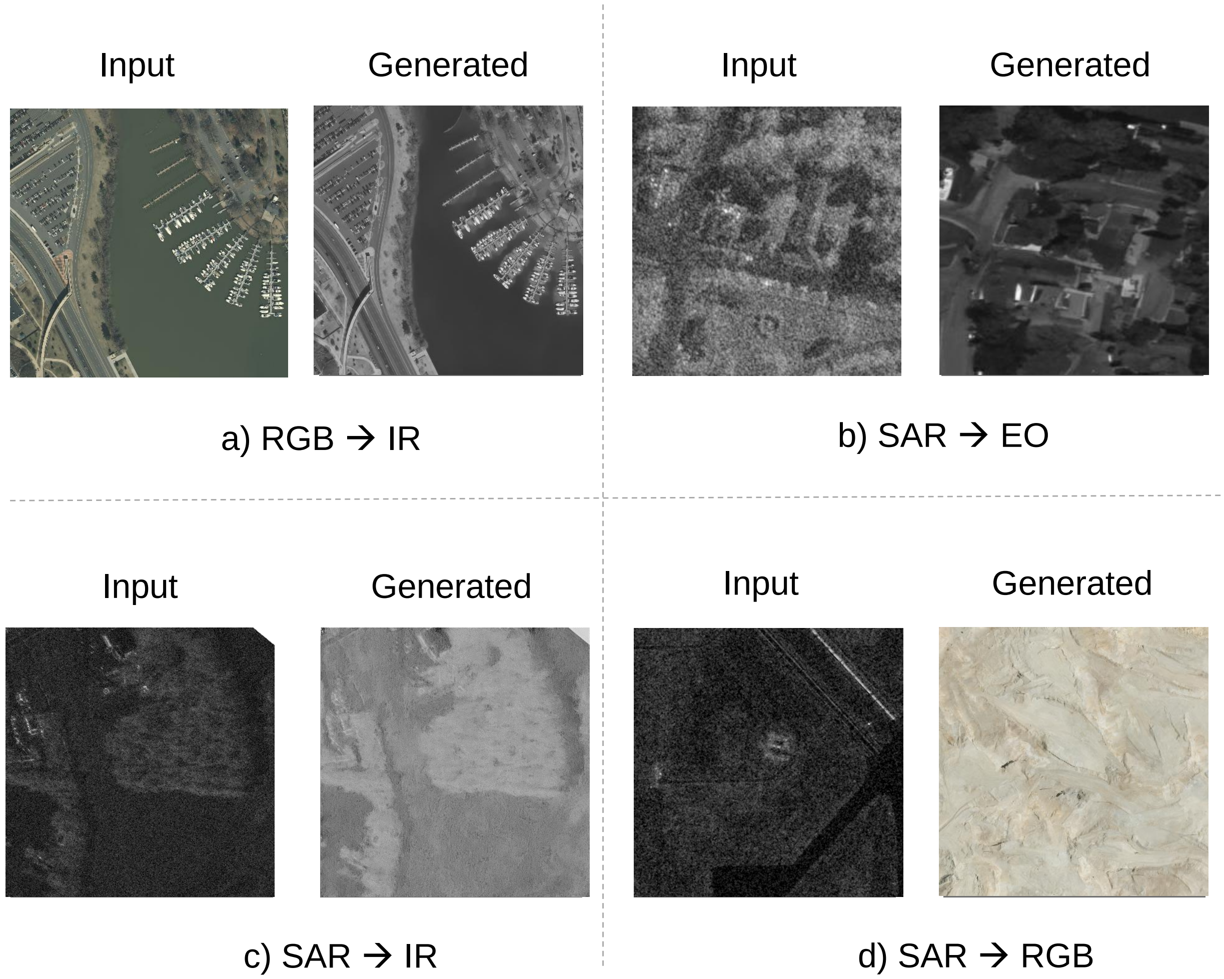


Figure 3. Qualitative results across all four MAVIC-T translation tasks. Each row corresponds to one task (SAR→EO, SAR→RGB, SAR→IR, RGB→IR). Columns show: source image, EarthBridge output, and ground-truth target. Our model preserves structural layout from the source while synthesizing faithful target-modality textures across both 256×256 and 1024×1024 resolutions.

follows the weighted Karras bridge loss with task-specific batch sizes and augmentations.

7. Lessons and Insights

We summarize key lessons and insights from developing EarthBridge for the MAVIC-T translation track.

Insight 1: Diffusion Bridge vs. Standard Diffusion

The diffusion bridge directly builds a mapping from the source domain to the target domain, whereas standard diffusion starts from a noise prior (typically Gaussian noise) and requires a conditional generation strategy to accomplish this task. We hypothesize that the diffusion bridge is more efficient for training and achieves better results for cross-modal translation. However, standard diffusion can be more flexible for further conditional development, e.g., leveraging pre-trained models for a good prior. There is a trade-off between these approaches; we encourage readers to further explore both directions.

Insight 2: Resolution Scaling and Multi-Stage Curriculum

For 1024px modeling, we train on 512px only and directly infer on 1024px, which yields strong results while saving training compute budget. If time permits, we may further perform a quick fine-tuning on 1024px directly—we term this approach *multi-stage resolution curriculum learning*.

Insight 3: Task Difficulty and Input Modality

We found that SAR→EO and RGB→IR are substantially easier than the other two tasks. For RGB→IR, the three-channel input provides sufficient information for the translation. For SAR→EO, the relative ease may stem from both modalities originating from the same sensor, reducing the domain gap.

Insight 4: Inference Steps and Metric Trade-offs

Using the DBIM ODE solver, few-step inference yields visually appealing results. For RGB→IR, even a single-step generation is sufficient, and increasing steps tends to hallucinate details. In practice, however, many more inference steps greatly improve FID and LPIPS in the competition metrics, even when L1 remains unchanged—highlighting a trade-off between perceptual quality, metric optimization, and inference efficiency.

8. Lessons and Insights

We summarize key lessons and insights from developing EarthBridge for the MAVIC-T translation track.

9. Conclusion

In this paper, we presented EarthBridge, a novel solution for the 4th Multi-modal Aerial View Image Challenge – Translation (MAVIC-T). By leveraging Diffusion Bridge Implicit Models (DBIM), we generalized the diffusion bridge framework with non-Markovian processes, enabling efficient and deterministic cross-modal translation. Our approach effectively handles disparate sensor characteristics across EO, IR, and SAR modalities through pixel-level modeling and a specialized training protocol featuring Karras-weighted bridge scalings. Quantitative and qualitative evaluations across four translation tasks demonstrate EarthBridge's superior ability to preserve spatial structure while generating high-fidelity target textures. Future work will explore the integration of adaptive domain shift mechanisms and further optimization for real-time remote sensing applications. **[TODO: Add final ranking and summary of key findings.]**

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Supplementary Material